One-loop torus valleys: local confinement, periodicity and spectral limits

Sector-gap result. For the explicitly specified flat-holonomy benchmark $H_{\mathrm{hol}} = (\hbar c/L)[-(g^2/2)\Delta_b + 2\Phi(b)/\pi^2]$ on the 4π torus, retaining centre transformations as physical symmetries, a centre-odd Weyl-even trial state gives

$$\Delta_{\mathrm{hol}} \le \frac{\hbar c}{L}\frac{32C_\psi}{\pi^2 v_0}\, g^{8/3} \exp\!\left[-\frac{\pi\sqrt{v_0}}{2g}\right]$$

for all sufficiently small $g > 0$. The fixed positive barrier v_0 and trial constant C_ψ are defined in Section 4. This rules out a uniform positive $g^{2/3}\hbar c/L$ lower bound for that benchmark's full Weyl-even spectrum. It does not identify the benchmark with the compressed field-theory Hamiltonian or prove its physical gap.

For a constant abelian $SU(2)$ background on T^3_L, the one-loop zero-point energy defined by discarding the formal $m = 0$ Fourier term in the Poisson-summed expression (distinct from the physical $k = 0$ oscillator) is

$$U(a) = \frac{2\hbar c}{\pi^2 L}\Phi(La), \qquad \Phi(b) = \sum_{m\in\mathbb{Z}^3\setminus\{0\}} \frac{1-\cos(m\cdot b)}{|m|^4}.$$

Here $[a] = L^{-1}$. This nonnegative bounded periodic function has the local expansion

$$U(a) = 2\hbar c|a| + \frac{C_M}{3\pi^2}\hbar c L|a|^2 + O(\hbar c L^2|a|^3).$$

The first term is the four transverse zero-mode oscillators' energy; the quadratic correction is useful input to a small-volume comparison. The formula is exact for this specified one-loop prescription, not for the interacting ground-state energy. A local Taylor expansion supplies neither the global operator bound nor the nonzero-mode estimates in the Feshbach reduction.

The derivation below retains the frequency count, Fourier normalization and symmetry argument. Weiss (*Phys. Rev. D* **24** (1981) 475, metadata) and Lüscher (1983, abstract) are historical source leads, not independent checks of this calculation. No novelty or full field-theory gap is claimed.

1. Frequencies in a constant abelian background

Let $a_i = \alpha_i T^3$, $i = 1, 2, 3$, with $\vec{\alpha} \in \mathbb{R}^3$, and write a for $\vec{\alpha}$ when no confusion arises. Decompose the adjoint fluctuation $\tilde{A}_i = \tilde{A}^3_i T^3 + \tilde{A}^+_i T^+ + \tilde{A}^-_i T^-$. The covariant derivative $D_j = \partial_j + [a_j, \cdot]$ acts on the neutral component as ∂_j and on the charged components as $\partial_j \pm i\alpha_j$, so on the Fourier mode $e^{ik\cdot x}$, $k \in (2\pi/L)\mathbb{Z}^3$, the charged components have covariant momentum $k \pm a$. The quadratic form $\frac{1}{2g^2}\int |D_i\tilde{A}_j - D_j\tilde{A}_i|^2$ is, for a constant abelian background, the free transverse form with k replaced by $k \pm a$ on the charged components, so after the usual transverse gauge choice the frequencies are

$$\hbar\omega^{(0)}_k = \hbar c|k| \quad \text{(neutral, two polarizations)},$$

$$\hbar\omega^{(\pm)}_k = \hbar c|k \pm a| \quad \text{(charged, two polarizations each)},$$

away from shifted zero frequencies. At $k = 0$ and $a \neq 0$, the background-covariant transverse condition $a \cdot \tilde{A}^{\pm} = 0$ leaves two charged spatial polarizations; the neutral components are the valley directions themselves: with $a = \alpha_1 T^3$ along x_1 only, these are the charged parts of $\tilde{A}_2, \tilde{A}_3$, four real oscillators of angular frequency $c|a|$, while the charged parts of $\tilde{A}_1$ are gauge directions removed by Gauss's law. Enhanced zero modes at $a = 0$ or shifted momenta require their own treatment; this local oscillator count is not a uniform projection across them.

The $k = 0$ term is C133. The SU(2) matrix model of C133 with $\vec{x}_1 = \lambda\hat{n}$, $\vec{x}_2 = \vec{x}_3 = 0$ has potential $\frac{1}{2}g^2(|\vec{x}_1 \times \vec{x}_2|^2 + |\vec{x}_1 \times \vec{x}_3|^2 + \cdots)$, whose transverse Hessian at that point gives the components of $\vec{x}_2, \vec{x}_3$ orthogonal to $\hat{n}$, four oscillators of frequency $g|\lambda|/\sqrt{m}$; in the field-theory variables of G07 Proposition 7 ($m = \hbar L^3/(g^2 c)$, $g_B^2 = \hbar c L^3/g^2$) this frequency is $c|a|$. The zero-point energy $4 \cdot \frac{1}{2}\hbar c|a| = 2\hbar c|a|$ is the confining term of G07 Theorem 6(1) and the floor of G08 Theorem 2 along the valley.

2. Poisson summation

The zero-point energy of the fluctuations relative to $a = 0$, summing two polarizations and both charges, is

$$U(a) = \frac{\hbar c}{2} \sum_{k \in (2\pi/L)\mathbb{Z}^3} \sum_{\text{pol}=1,2} \big(|k+a| + |k-a| - 2|k|\big) = \hbar c \sum_{k \in (2\pi/L)\mathbb{Z}^3} \big(|k+a| + |k-a| - 2|k|\big),$$

where the $k = 0$ term is $2\hbar c|a|$ and the neutral modes cancel. The raw summands are nonnegative and of order $|a|^2/|k|$; their unregularized sum diverges. Periodicity of a regulated sum alone does not justify exchanging this sum and subtraction. We use the specific Fourier finite-part prescription that discards the $m = 0$ term below. It restores periodicity by momentum reindexing and fixes the displayed one-loop potential; a different prescription needs its own matching. For $f(k) = |k|$ the distributional Fourier transform is $\hat{f}(x) = \int |k| e^{-ik\cdot x} d^3k = -8\pi/|x|^4$ for $x \neq 0$ (the formula $\widehat{|k|^\alpha} = 2^{\alpha+3}\pi^{3/2}\frac{\Gamma((3+\alpha)/2)}{\Gamma(-\alpha/2)}|x|^{-3-\alpha}$ at $\alpha = 1$, with $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$), and

$$\sum_{k \in (2\pi/L)\mathbb{Z}^3} f(k+a) = \Big(\frac{L}{2\pi}\Big)^3 \sum_{m \in \mathbb{Z}^3} \hat{f}(mL)\, e^{im\cdot La}$$

with the $m = 0$ symbol a prescription-dependent subtraction, not a separately evaluable distributional Fourier value. Hence

$$U(a) = \hbar c\Big(\frac{L}{2\pi}\Big)^3 \sum_{m \neq 0} \Big(-\frac{8\pi}{L^4|m|^4}\Big)\big(e^{im\cdot La} + e^{-im\cdot La} - 2\big) = \frac{2\hbar c}{\pi^2 L} \sum_{m \neq 0} \frac{1 - \cos(m \cdot La)}{|m|^4} = \frac{2\hbar c}{\pi^2 L}\Phi(La).$$

Matching a lattice dispersion to this continuum one-loop expression requires a separate cutoff estimate. No uniform relative $O(a_{\text{lat}}^2/L^2)$ error follows merely from the nonzero Fourier indices.

Properties of Φ: it is nonnegative, periodic with period 2π in each component, invariant under the cubic group, bounded by $2\sum' |m|^{-4}$, and zero exactly on $2\pi\mathbb{Z}^3$.

3. Expansion at the origin and the sign of the correction

Φ is not smooth at 0. Its Laplacian is the periodic Green's function $G(b) = \sum'_m e^{im\cdot b}|m|^{-2}$, which has the Coulomb singularity $2\pi^2/|b|$ at the origin (from $\int d^3m\, e^{im\cdot b}|m|^{-2}$) and a finite remainder: $G(b) = 2\pi^2/|b| + C_M + O(|b|^2)$, where C_M is the constant term of the periodic Coulomb Green's function of the cubic lattice, the analytically continued Epstein sum $\sum'_m |m|^{-2}$. Since $\Delta(\pi^2|b|) = 2\pi^2/|b|$, the function $\Phi - \pi^2|b|$ has Laplacian $C_M + O(|b|^2)$ near 0, and the only quadratic form with cubic symmetry and Laplacian C_M is $\frac{1}{6}C_M|b|^2$; so

$$\Phi(b) = \pi^2|b| + \tfrac{1}{6}C_M|b|^2 + O(|b|^3), \qquad U(a) = 2\hbar c|a| + \frac{C_M}{3\pi^2}\,\hbar c L|a|^2 + O(\hbar c L^2|a|^3).$$

The linear term $2\hbar c|a|$ is the $k = 0$ contribution of Section 1, so the nonzero modes contribute

$$U_\perp(a) = U(a) - 2\hbar c|a| = \frac{C_M}{3\pi^2}\hbar c L|a|^2 + O(\hbar c L^2|a|^3),$$

with $C_M < 0$ for the cubic lattice (lattice-sum tables; the sign is not re-derived here and affects only the direction of a subleading correction). In the rescaled variables $a = g^{2/3}\xi/L$ this is

$\frac{C_M}{3\pi^2} g^{4/3}|\xi|^2\,\hbar c/L$, of relative order $g^{2/3}$ to the zero-mode Hamiltonian $g^{2/3}h_3\,\hbar c/L$: the size that (H3) of the formal local power counting suggests for the Feshbach reduction. The coefficient is identified for this one-loop prescription, not for the full compressed-operator gap hypothesis (H3).

3a. General constant backgrounds at quadratic order

The following extension is conditional: suppose a specified gauge-fixed nonzero-mode determinant, including its longitudinal and ghost contributions, defines a smooth even gauge-invariant one-loop function near the trivial background and is matched to the same Fourier prescription as the abelian calculation. An unfixed Hessian at a general noncommuting off-shell background does not automatically have these properties. Under that assumption its quadratic term is constrained by symmetry. It is invariant under the constant gauge transformations, which rotate the three colour vectors $\vec{a}_i$ simultaneously, under the cubic group acting on the spatial index, and under $a \to -a$. The quadratic invariants of $(\vec{a}_1, \vec{a}_2, \vec{a}_3)$ under $SO(3)_{\text{colour}}$ are the bilinears $\vec{a}_i \cdot \vec{a}_j$, and the only cubic-invariant symmetric form in the spatial indices is δ_{ij}, so

$$U_\perp(a) = c\,\hbar c L \sum_{i=1}^{3} |\vec{a}_i|^2 + O(\hbar c L^2 |a|^3),$$

with a single pure number c, which the abelian valley ($\vec{a}_i = \alpha_i \hat{e}_3$, $\sum_i |\vec{a}_i|^2 = |\alpha|^2$) determines: $c = C_M/(3\pi^2)$. The Landau-level structure of non-commuting backgrounds cannot change this quadratic invariant within the same smooth one-loop prescription. The remainder is at least cubic in this local estimate. This identifies a candidate coefficient for (H3); symmetry does not establish a cutoff-uniform expansion, a global relative bound or the gap of the compressed operator.

3b. The abelian theory in the same expansion

For compact $U(1)$ the constant modes commute, the quartic potential vanishes, and the photons are neutral, so the one-loop potential of the holonomy is identically zero: the constant-mode Hamiltonian is the free Laplacian on the torus of holonomies, $H_0 = \frac{e^2 c}{2\hbar L^3}|p|^2$ with a periodic of period $2\pi/L$, spectrum $e^2\hbar c|n|^2/(2L)$, $n \in \mathbb{Z}^3$, gap $e^2\hbar c/(2L)$. The abelian small-volume gap is linear in e^2, not $e^{2/3}$, because there is no commutator to confine the holonomy and no charged mode to give it a potential; with e fixed by the absence of running, the $1/L$ decay is uncompensated and the infinite-volume theory is gapless, in agreement with the Coulomb phase of Guth and Fröhlich–Spencer. In the nonabelian matrix model, the $g^{2/3}$ scale comes from the kinetic/quartic commutator balance. Running of the field-theory coupling requires a general-background renormalization and matching calculation; it cannot be read from the commuting-valley function U alone.

4. Saturation and central holonomies

The full one-loop potential satisfies $0 \le U \le 2\hbar c \max \Phi/(\pi^2 L) = O(\hbar c/L)$, and its minima are at $La \in 2\pi\mathbb{Z}^3$. For $SU(2)$ with adjoint charges ± 1, $La = 2\pi e_i$ gives fundamental holonomy $\exp(2\pi i T^3) = -1$. With strictly periodic gauge transformations these are distinct central holonomy configurations. Strictly periodic $SU(2)$ gauge transformations identify covering shifts by $4\pi/L$ (and Weyl reflection), so the infinitely many covering minima represent 2^3 central holonomy triples. If transformations periodic only up to the centre are instead gauged, the identification and allowed sectors change. A configuration is not itself a quantum state.

When centre transformations are global symmetries, their characters on three torus cycles label electric-flux sectors: $\mathbb{Z}_N^3$ gives 8 labels for $SU(2)$ and 27 for $SU(3)$. The latter has three central holonomies per direction. These labels and the one-loop barrier scale $O(\hbar c/L)$ determine

neither spectral splittings nor spectral weights. Tunnelling and electric-flux interpretations require the corresponding Hamiltonian, boundary conventions and estimates; they are not derived by counting minima.

In the isolated matrix model, balancing kinetic and quartic terms gives $La \sim g^{2/3}$ and the energy scale $g^{2/3}\hbar c/L$. Its linear valley term is the local limit of U, whereas U itself saturates and is periodic. Crucially, $U_\perp = U - 2\hbar c|a|$ is **not** bounded or periodic on the covering space: the subtracted linear term grows. The boundedness of U therefore cannot establish a global perturbation bound for $U_\perp$ relative to the confining matrix operator. Linear confinement may control $|a|$ and allow some form bound; the small relative bound needed for a gap comparison has not been supplied. Nor does a local quadratic estimate control arbitrarily large $|\xi|$ after $a = g^{2/3}\xi/L$.

Localization near one minimum and an expansion of the full field-theory gap require the Feshbach hypotheses and a global treatment of the holonomy domain. The informal comparison $g(L)^{2/3} \sim 1$ marks the loss of a small parameter; it proves no delocalization, sector coalescence or crossover at a specified value of $z = \Delta L/(\hbar c)$.

A centre-odd variational countertest

The distinction can be proved without calculating a tunnelling asymptotic. Take the declared scalar operator

$$h_g = -\frac{g^2}{2}\Delta_b + W(b), \qquad W(b) = \frac{2}{\pi^2}\Phi(b), \qquad b \in \mathbb{T}^3_{4\pi} = (\mathbb{R}/4\pi\mathbb{Z})^3,$$

on flat $L^2(d^3b)$ with its periodic H^2 domain, and set $H_{\mathrm{hol}} = (\hbar c/L)h_g$. Its Weyl-even subspace consists of $f(-b) = f(b)$. Centre translations are $(T_i f)(b) = f(b+2\pi e_i)$, not gauge identifications in this model. These commuting involutions commute with h_g and with Weyl reflection. The nonnegative bounded continuous W has exactly eight zeros $Z = (2\pi\mathbb{Z})^3 \bmod 4\pi$, and $W(b) \le C_W|b|$ near zero. No nonzero-mode, orbit-measure or Berry term is included in this benchmark; matching those terms is a separate field-theory obligation. Rayleigh quotients below use the closed quadratic form $q_g[v] = (g^2/2)\int|\nabla v|^2 + \int W|v|^2$, with domain H^1; the Lipschitz cutoffs need not belong to the operator's H^2 domain.

Let $e_0(g)$ and $\Omega_g > 0$ be the normalized ground eigenpair. Compactness and positivity improvement give a simple ground state, fixed by every T_i and by Weyl reflection. Its energy is also the lowest energy in the Weyl-even subspace. Fix a normalized $\psi \in C_c^\infty(B_1(0))$, and put

$$C_\psi = \frac{1}{2}\int|\nabla\psi|^2 d^3z + C_W\int|z|\,|\psi(z)|^2 d^3z.$$

For $s = g^{2/3}$ small enough that $W(b) \le C_W|b|$ on B_s, the normalized trial $s^{-3/2}\psi(b/s)$ gives

$$0 \le e_0(g) \le C_\psi g^{2/3}. \tag{V1}$$

Indeed its kinetic energy is $g^2 s^{-2}\int|\nabla\psi|^2/2$ and its potential energy is at most $C_W s\int|z||\psi|^2$. This is a ground-energy upper bound, not an excitation-gap estimate.

Use the flat torus distance, let A be the union of the eight closed balls of radius $r = \pi/4$ about Z, and define the fixed constant

$$v_0 = \min_{\mathrm{dist}(b,Z)\ge\pi/4} W(b) > 0.$$

Assume g is small enough for (V1) and $e_0 \le v_0/4$. The Lipschitz weight

$$\rho(b) = \frac{\sqrt{v_0}}{2}\,\mathrm{dist}(b, A)$$

vanishes on A. The weighted eigenfunction identity, with kinetic coefficient $g^2/2$ and weight ρ/g, reads

$$\frac{g^2}{2}\int |\nabla(e^{\rho/g}\Omega_g)|^2 + \int \left(W - e_0 - \frac{|\nabla\rho|^2}{2}\right) e^{2\rho/g}\Omega_g^2 = 0.$$

On A^c, its second coefficient is at least $v_0 - v_0/4 - v_0/8 \geq v_0/2$; on A it is at least $-e_0$. Consequently

$$\int_{A^c} e^{2\rho/g}\Omega_g^2 \leq \frac{2e_0}{v_0}. \tag{V2}$$

All constants here are fixed by this compact benchmark, not uniform field-theory or continuum estimates.

Choose a real periodic Lipschitz $f = f(b_1)$ with plateaus $+1$ about $b_1 = 0$ and -1 about $b_1 = 2\pi$. Interpolate linearly across the two strips of half-width $\delta = \pi/4$ centered at $\pi, 3\pi$. Then $f(b_1 + 2\pi) = -f(b_1)$, $f(-b_1) = f(b_1)$, $|f'| \leq 4/\pi$, and $f^2 = 1$ off their union B. Every point of B has distance at least $\pi/2$ from A, so (V2) gives

$$p_B := \int_B \Omega_g^2 \leq \frac{2e_0}{v_0} e^{-\pi\sqrt{v_0}/(2g)} \leq \frac{1}{2}.$$

The vector $f\Omega_g$ is Weyl-even and has centre characters $(-,+,+)$. It is orthogonal to Ω_g by the T_1 symmetry, and the exact ground-state transform gives

$$\frac{q_g[f\Omega_g] - e_0\|f\Omega_g\|^2}{\|f\Omega_g\|^2} = \frac{g^2}{2}\frac{\int |\nabla f|^2\Omega_g^2}{\int f^2\Omega_g^2} \leq \frac{8g^2}{\pi^2}\frac{p_B}{1-p_B} \leq \frac{32g^2e_0}{\pi^2 v_0} e^{-\pi\sqrt{v_0}/(2g)}.$$

Rayleigh–Ritz and (V1) prove the displayed physical bound for the gap above the vacuum in the full Weyl-even benchmark. They also upper-bound the lowest $(-,+,+)$ sector energy above E_0. In particular

$$\frac{L\Delta_{\mathrm{hol}}}{\hbar c\, g^{2/3}} \leq \frac{32C_\psi}{\pi^2 v_0} g^2 e^{-\pi\sqrt{v_0}/(2g)} \longrightarrow 0.$$

This is an upper bound, not a matching tunnelling formula. □

Even cubic symmetry alone does not remove this benchmark trial. The three vectors $f(b_i)\Omega_g$ have different centre characters and are mutually orthogonal, also for $q_g - e_0$. Cubic symmetry gives them equal norms and form energies. Their sum is invariant under coordinate permutations and sign changes and has the same Rayleigh quotient. Thus the bound also holds in the cubic-invariant subspace when all centre characters are admitted. This sum has no trivial-centre component, although it has no single charged character. The calculation concerns the declared scalar symmetry action, not an imported classification of the full gauge-theory levels.

The trial is excluded if centre translations are gauged or the Hilbert space is restricted to the trivial centre character. That exclusion alone proves no $g^{2/3}$ lower bound in the remaining space. Thus H3 must name its sector, global holonomy domain, measure and projection; a local matrix spacing cannot stand for every centre sector. Nor does this finite-volume benchmark settle any infinite-volume gap: $g(L)$, the full operator and the order of limits must first be identified. The vacuum subspace cyclic for a centre-neutral local observable algebra must also be distinguished from the full finite-box Gauss-invariant space; the charged trial is not produced by a neutral multiplication observable.

Source boundary. The separation of channel energies and electric-flux splittings is established literature, not a novelty claim. Lüscher–Münster's 1984 publisher abstract defines its z using the $J^p = 0^+$ sector. Weisz–Ziemann, *Nucl. Phys. B* **284** (1987) 157–170, printed p. 168, passage-read, discusses electric-flux sectors, perturbative degeneracy and tunnelling. The Koller–van Baal calculation cited there has not been read here. The bound above is a written derivation for the declared scalar benchmark, not a verification of that field-theory calculation.

5. Consequence for STATE

The explicit one-loop input is a periodic bounded U with central minima and a local correction $U_\perp = (C_M/(3\pi^2))\hbar c L|a|^2 + O(\hbar c L^2|a|^3)$. The symmetry argument identifies its quadratic form under the stated smooth-prescription assumption. Global (H3), nonzero-mode comparison, $SU(3)$ matrix control, cutoff removal and physical spectral transport remain separate. The centre-odd variational bound additionally shows why the global benchmark gap differs from a local matrix spacing; H3 must specify its centre sector. The abelian comparison alone proves no nonabelian continuum gap.